Transfer: gravity and motion

antocciphysics | TEACHER KEY | CPA

Learning target: Apply gravity, projectile and orbital reasoning to a new set of cases.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 13; 3 course days later.*
Velocity tangent; acceleration inward at both positions. Constant speed does not mean constant velocity.

02. Explain a worked example (worked)
Both axes evolve over the same 2 s; horizontal distance 8 m, vertical motion governed by gravity.

03. Look and explain (model)
Tangent v, inward gravity; shared freefall removes support force.

04. Build the solution (practice)
12 m; new 6 m.

05. Use the idea (reason)
Need time phase/geometry/Earth position/uncertainty; valid specific request.

06. Choose a method (mixed) *Review: Unit 2, Day 4; 12 course days later.*
4 N; divide by 36 because force follows inverse square separation.

07. Exit check (exit)
g 2 N/kg; force 6 N inward.

08. STRETCH: Apply it another way (stretch)
Triple horizontal speed; keep height, initial vertical velocity and gravitational field fixed, with negligible air resistance.

Teaching note: This is a self-contained daily transfer worksheet, with supplied practice data. Keep any separately assigned formal assessment independent.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 13; 3 course days later.*
Speed becomes 1/3; period becomes 27 times as large. v∝r^(-1/2), T∝r^(3/2).

02. Explain a worked example (worked)
Both axes evolve over the same 2 s; horizontal distance 8 m, vertical motion governed by gravity.

03. Represent and interpret (model)
Tangent v, inward gravity; shared freefall removes support force.

04. Solve and justify (practice)
t 2 s; range 12 m; vy=-20 m/s; speed√436=20.9 m/s.

05. Analyze the reasoning (reason)
Need time phase/geometry/Earth position/uncertainty; valid specific request.

06. Choose a method (mixed) *Review: Unit 2, Day 4; 12 course days later.*
Fixed masses: 4 r. Both masses enlarged: 24 r gives the required force fraction.

07. Exit check (exit)
v 4 m/s; a=.64 m/s²; T 39.3 s; direction changes while speed stays constant.

08. STRETCH: Apply it another way (stretch)
t=3 s, range 36 m, velocity (12,-30) m/s. At half g, time and range multiply by √2; new range≈50.9 m.

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